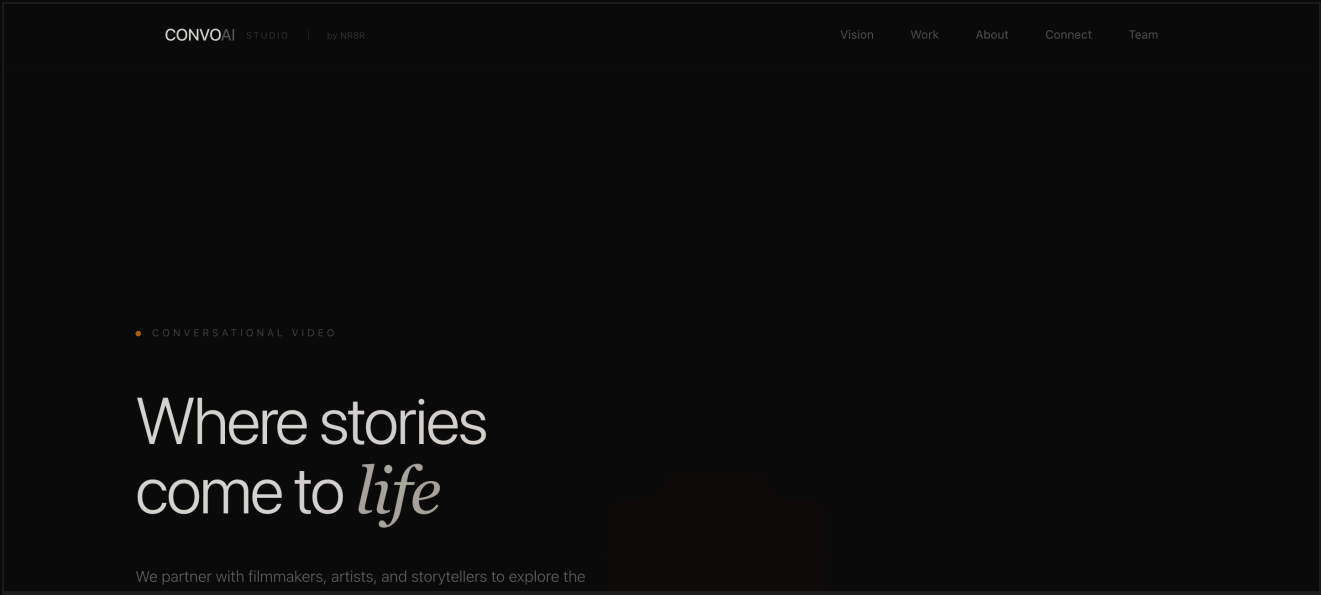


ConvoAI Studio

An internal conversational video demo platform for NR8R - with a clinical chest-pain intake pipeline from live Tavus conversations to physician-ready PDFs.



Dev Ganugapenta

Founder, InitDev

Stack

Next.js 16 · Supabase · Tavus · Claude · Resend

Brand

NR8R (Narrator Studio)

Evidence date

2026-07-17 · Local verification

1. Executive assessment

ConvoAI Studio is a proprietary NR8R control plane for configuring Tavus AI-replica demos, issuing time-boxed shareable links, and - in the Flo intake path - turning multimodal interviews into structured clinical documentation.

What it is Internal / agency delivery tool: admin dashboard + branded public demo viewer + post-call intake reporting. Not a self-serve multi-tenant SaaS signup product. IMPLEMENTED	Maturity Production-oriented: CI lint/typecheck/build, webhook redundancy, clinical PDF iteration in git history. Automated product tests: none. Lint currently fails (35 issues). PARTIAL ops maturity
Audience NR8R admins; external demo recipients; clinicians receiving Flo PDFs via <code>report_recipient</code>.	Why it matters Compresses demo ops and clinical intake documentation into one owned pipeline - video conversation, structured extraction, PDF, email - instead of a thin Tavus wrapper.
Commercial status: UNVERIFIED EXTERNAL CLAIM Revenue, customer volume, and clinical deployment outcomes are not evidenced in the repository. README "award-winning" language is marketing context only.	

2. Verified proof at a glance

14

API ROUTE FILES

22

HTTP HANDLERS

8

APP PAGES

3

DB TABLES + RLS

42

GIT COMMITS

0

AUTOMATED TESTS

PASS

TSC + BUILD

FAIL

ESLINT (35)

FACT	EVIDENCE	LABEL
Typed Tavus API client (~672 LOC)	lib/tavus-client.ts	IMPLEMENTED
Chest-pain structured extraction via Claude	lib/transcript-analyzer.ts	IMPLEMENTED
Clinical PDF + Resend delivery + dedup	lib/pdf-generator.tsx, intake-report.ts	IMPLEMENTED
Webhook + client complete + sendBeacon	webhooks/tavus, demo/complete, branded viewer	IMPLEMENTED
Perception analysis in automated PDF path	Stored; not passed in generateIntakeReport	PARTIAL
Admin send-report authz	app/api/admin/send-report/route.ts	GAP
Unit / e2e suite	No *.test.* / Playwright product tests	PROPOSED

3. Product and business context

Problem

Agencies and clinical partners need lifelike AI video demos that are configurable, brandable, and shareable - and, for Flo, need the interview to produce physician-usable structured notes without manual transcription.

Primary workflows

1. Admin creates a project binding Tavus `persona_id` + `replica_id`, greeting/context, branding, session duration.
2. Admin generates a 24-hour `demo_sessions` link.
3. Recipient opens branded viewer; optionally completes intake form (patient name + doctor email).
4. Live Tavus conversation runs in `iframe`; session moves `pending` → `active` → `completed`.
5. On Flo path: transcript → Claude clinical schema → PDF → email to doctor, with `report_sent_at` dedup.

Value (technical, evidenced)

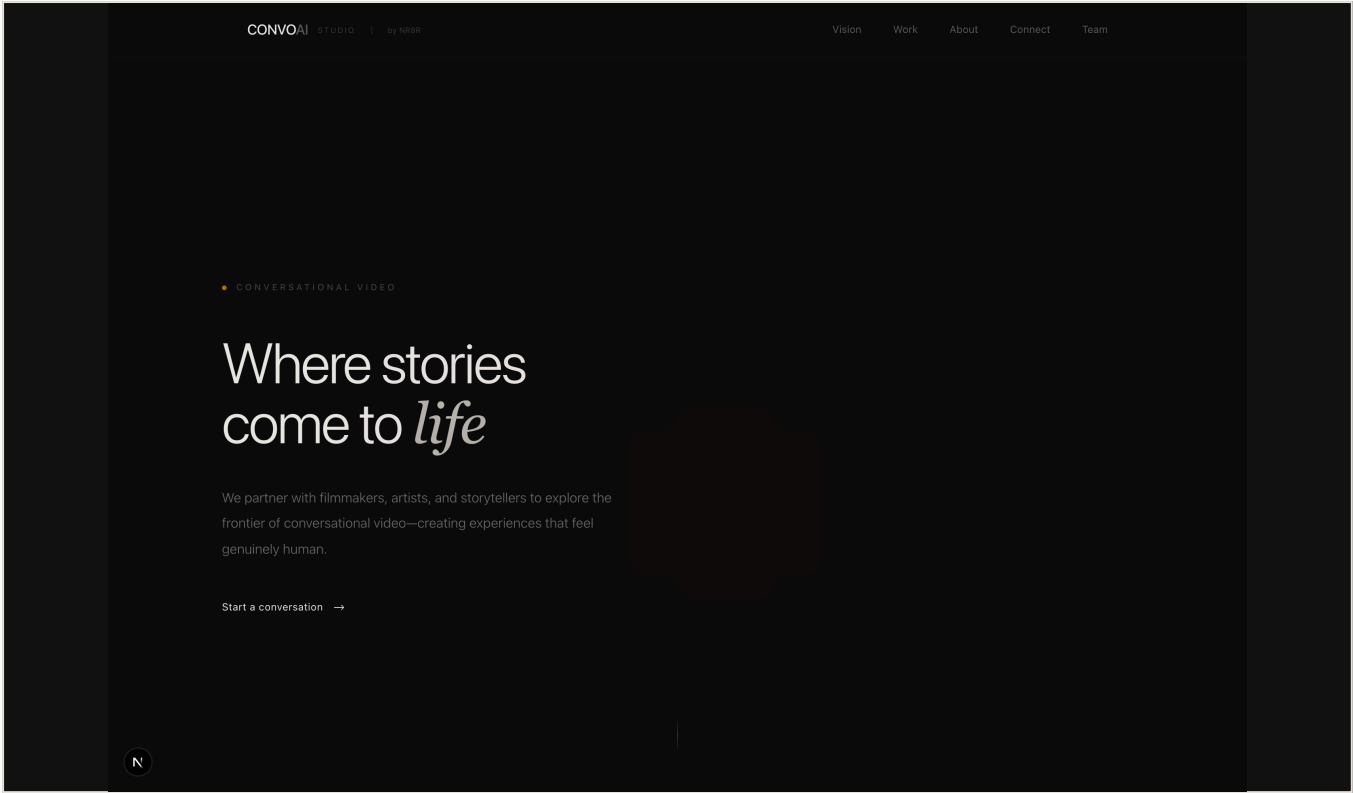
- Demo ops speed: reusable project templates + one-click share links.
- White-label delivery: per-project colors, logos, CTAs, welcome copy.
- Clinical documentation leverage: domain-encoded CAD/AOD/PE risk + ROS extraction.
- Reliability under partial failure: multiple completion triggers for transcript readiness.

Defensibility is primarily workflow + domain encoding + operational packaging of Tavus - not proprietary model training.

INFERRED

4. Live product walkthrough

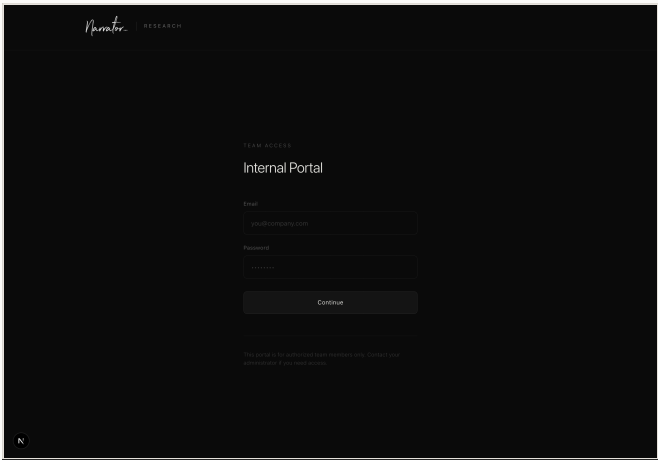
All screenshots: local Next.js · Playwright Chromium · 1440×1000 or 390×844 · DPR 2 · 2026-07-17. No production customer data.



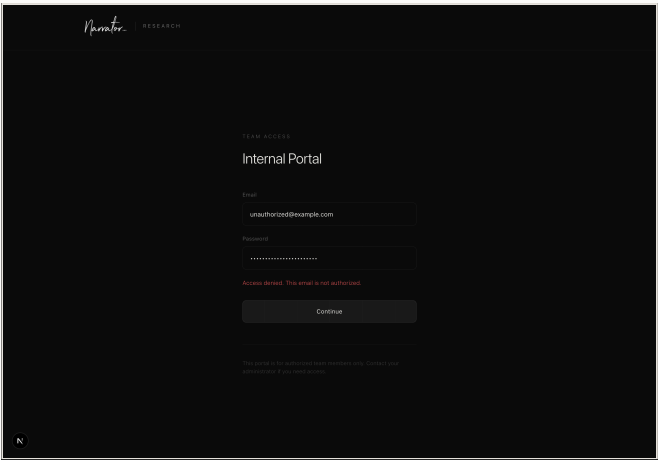
Public landing hero - ConvoAI Studio by NR8R brand composition.

IMPLEMENTED · Local development capture · Playwright

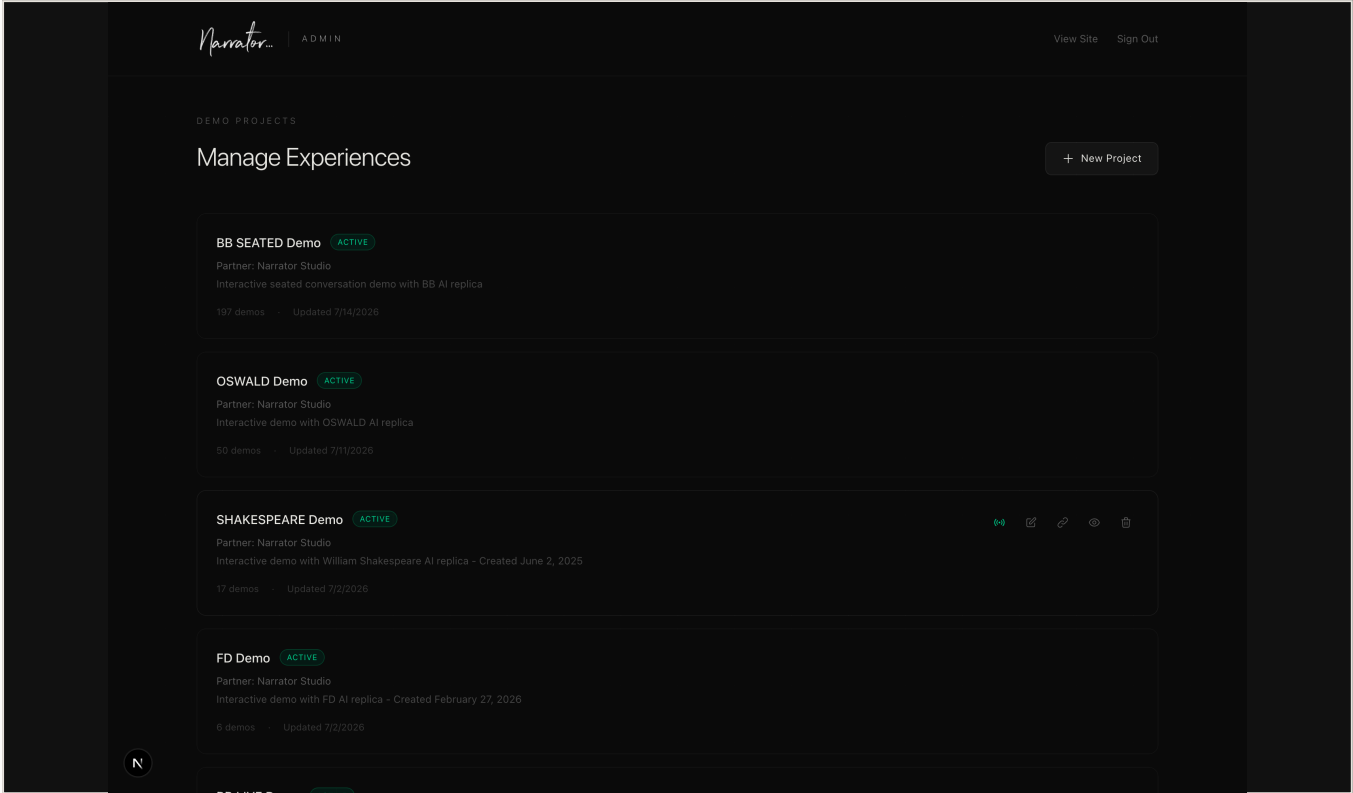
4. Live product walkthrough (continued)



Team login - allowlist admin entry. **IMPLEMENTED**

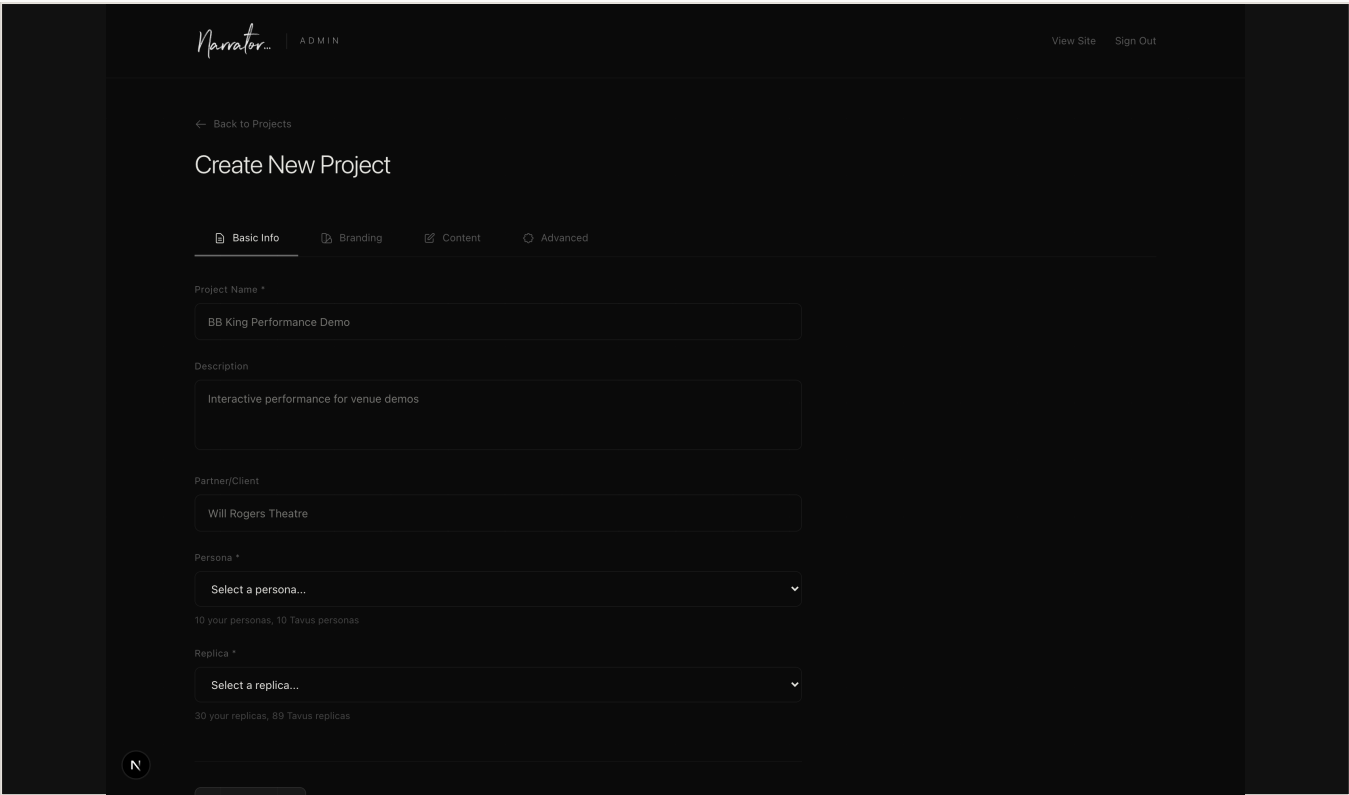


Auth failure state - unauthorized email rejected. **IMPLEMENTED**



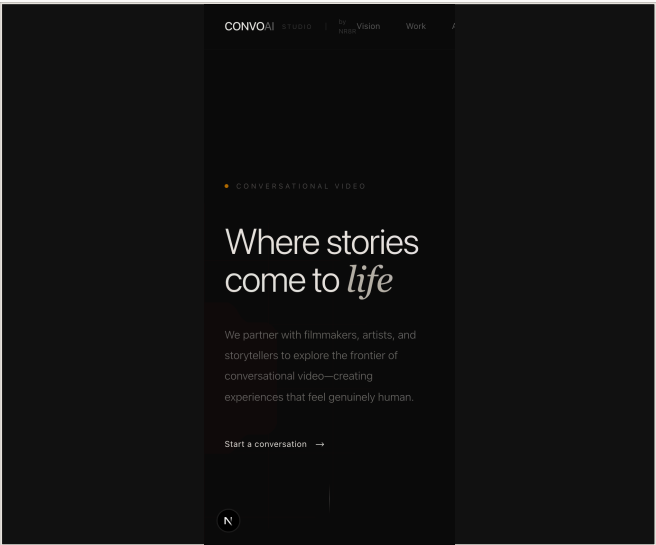
Authenticated admin dashboard - project control plane for demos. **IMPLEMENTED** · Local development capture · Playwright

4. Live product walkthrough (continued)

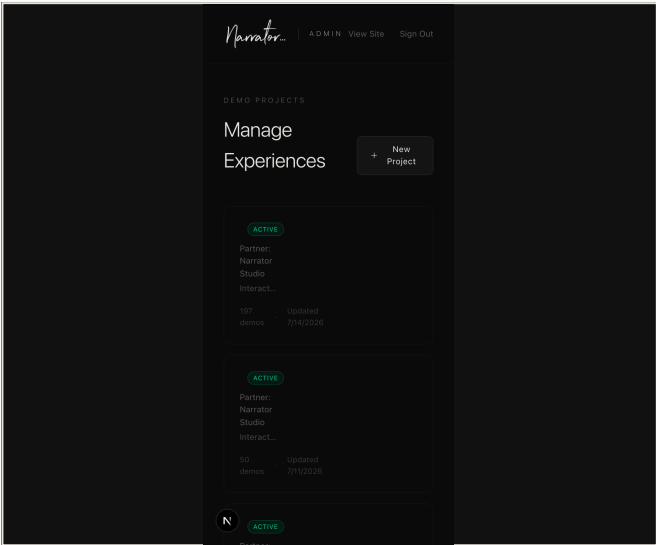


Project create/edit surface - persona/replica and branding configuration.

IMPLEMENTED · Local development capture · Playwright

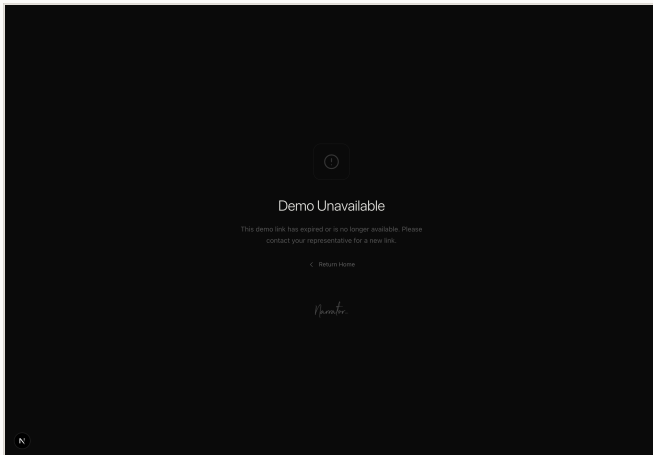


Mobile landing (390×844). **IMPLEMENTED**

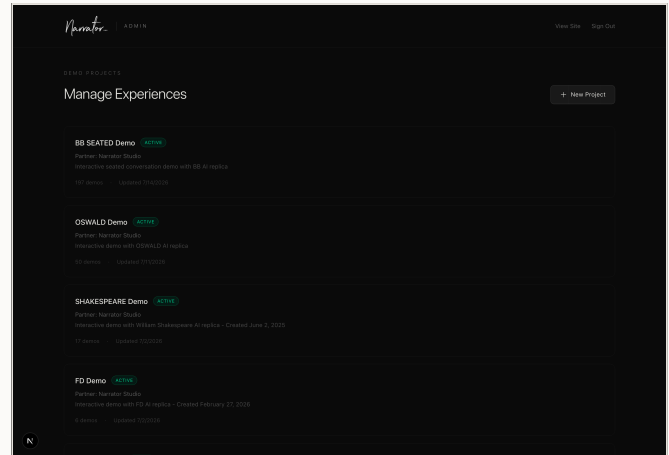


Mobile dashboard. **IMPLEMENTED**

4. Live product walkthrough (continued)

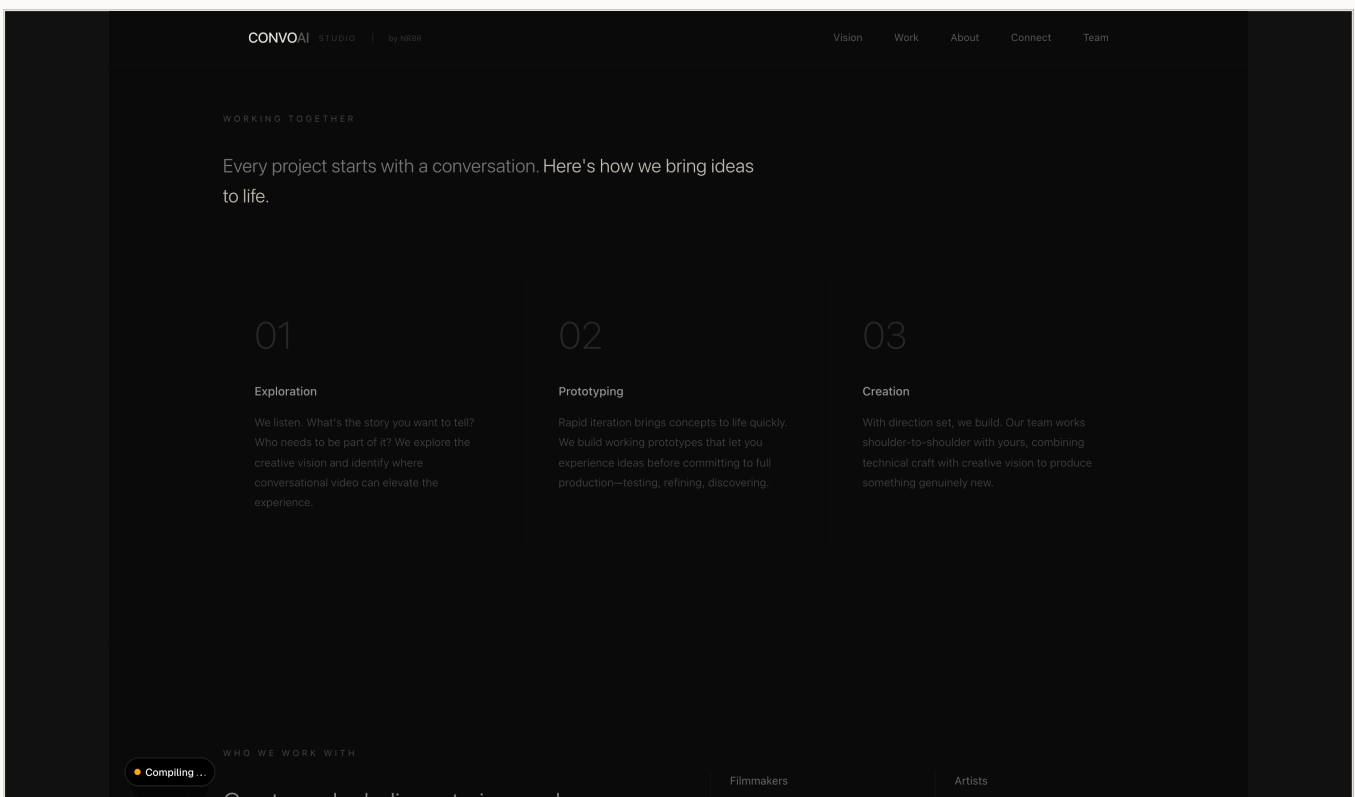


Invalid demo session - failure path. **IMPLEMENTED**



/studio redirect to dashboard. **DEPRECATED PATH**

Intentionally omitted: live Tavus clinical calls and emailed PHI PDFs (safety). Those capabilities are evidenced in code: `lib/intake-report.ts`, `lib/transcript-analyzer.ts`, `app/api/webhooks/tavus/route.ts`.



Landing work/capability narrative section.

IMPLEMENTED · Local development capture · Playwright

5. Platform capability map

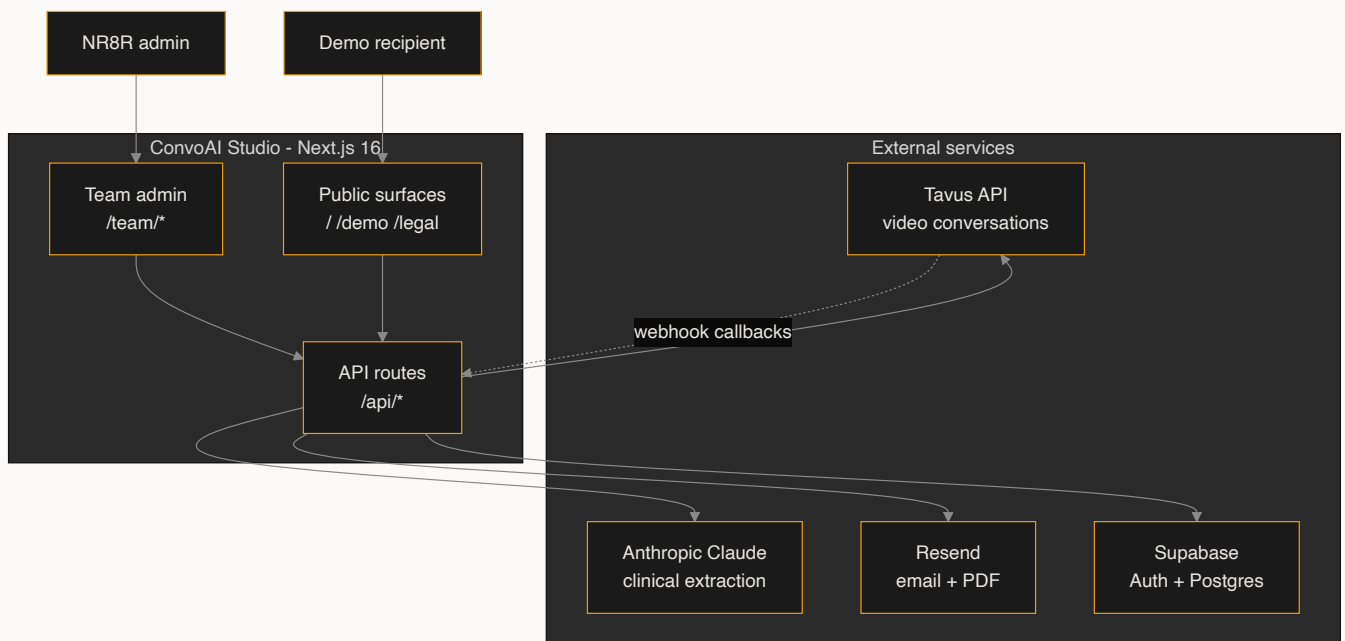
SURFACE	PROVES	STATUS
Public landing /	Brand positioning, NR8R narrative	IMPLEMENTED
/team/login + cookie auth	Privileged operator boundary	IMPLEMENTED
/team/dashboard	Project CRUD, branding, demo links	IMPLEMENTED
/demo/[sessionId]	Branded viewer + intake gate + iframe	IMPLEMENTED
Demo APIs start/intake/complete	Session lifecycle + report trigger	IMPLEMENTED
Tavus webhook	Async completion + transcript fetch	IMPLEMENTED
Intake report pipeline	Claude → PDF → Resend	IMPLEMENTED
Perception in auto PDF	Multimodal observation rendering	PARTIAL
Personas/replicas/conversations APIs	Tavus resource management	IMPLEMENTED
Billing / multi-tenant SaaS	-	PROPOSED
Automated test suite	-	PROPOSED

6. Complete technology stack

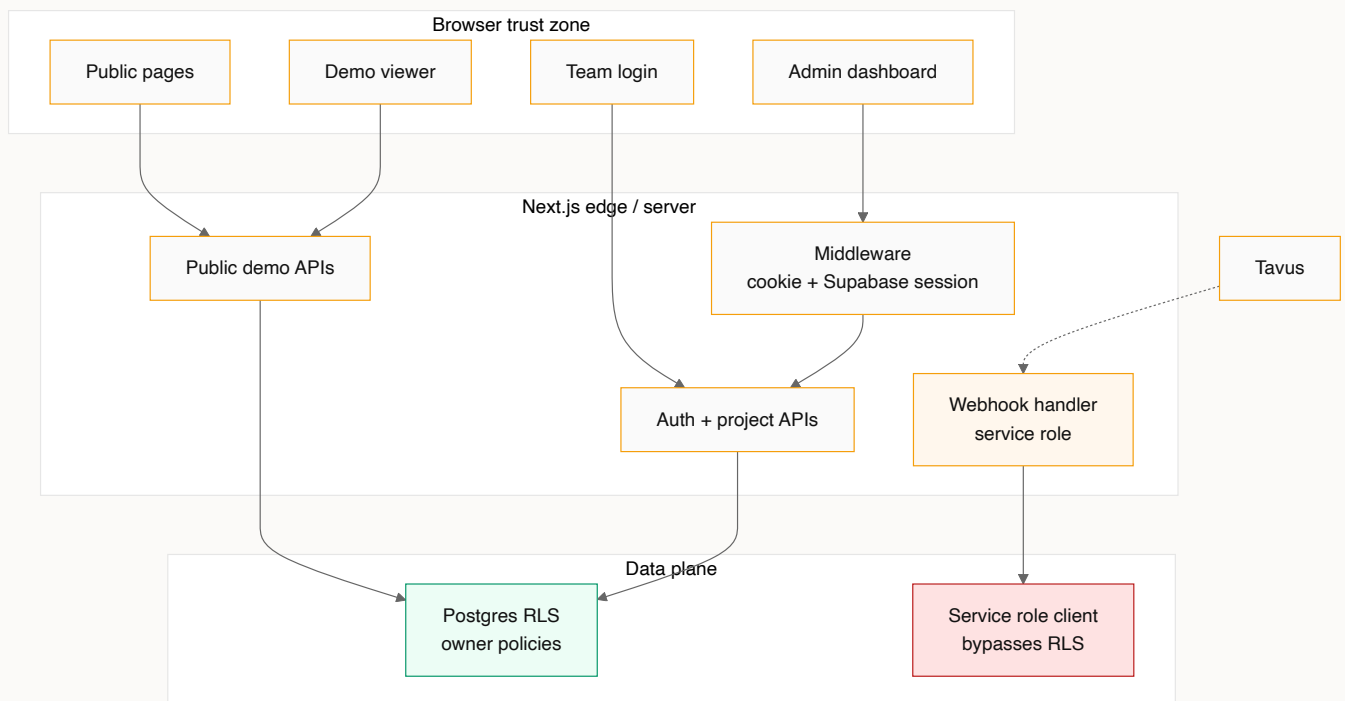
TECHNOLOGY	HOW IT IS USED	MATURITY
Next.js 16 App Router	Pages, API routes, middleware, RSC defaults	Build PASS
React 19 + TypeScript	Dashboard + branded viewer clients	tsc PASS
Tailwind CSS 4	Dark editorial UI; amber accent	IMPLEMENTED
Supabase Postgres + Auth	projects, demo_sessions, RLS, service role for webhooks	IMPLEMENTED
Tavus API v2	Conversations, personas, replicas via typed client	IMPLEMENTED
Anthropic Claude	<code>c\laude-sonnet-4-20250514</code> chest-pain JSON extraction	IMPLEMENTED
@react-pdf/renderer	Clinical intake PDF buffer	IMPLEMENTED
Resend	Doctor email with PDF attachment	IMPLEMENTED
GitHub Actions CI	lint, tsc, build jobs	IMPLEMENTED (lint currently red locally)
Radix Slot + CVA	Minimal shared button/badge primitives	IMPLEMENTED

Installed packages alone are not proof - usage verified via imports in `lib/*` and API routes.

7. System architecture

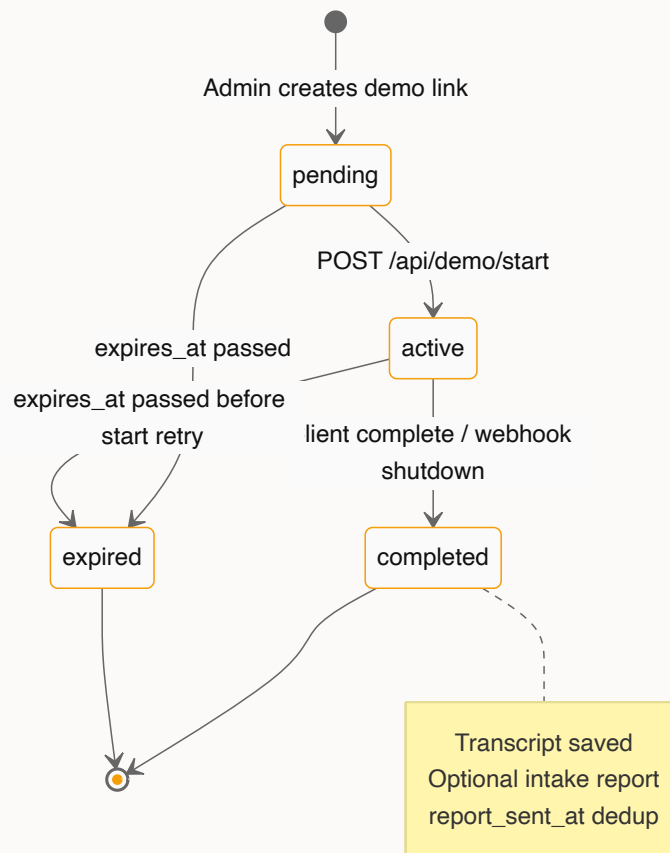


System context - ConvoAI Studio between admins/prospects and Tavus, Claude, Resend, Supabase. Solid = verified.



Trust boundaries - middleware-protected admin APIs vs public demo APIs vs service-role webhook path.

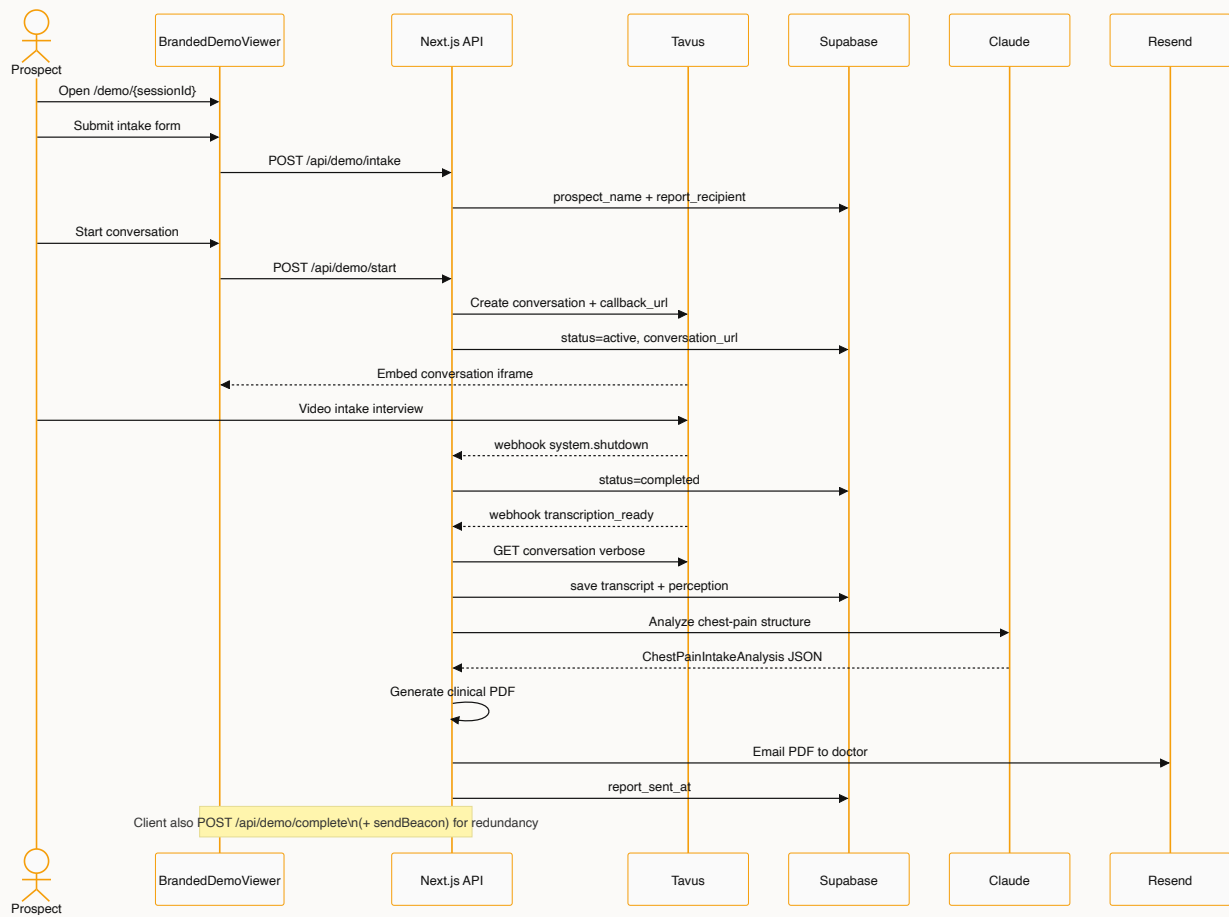
8. Critical end-to-end workflows



Demo session state machine: pending → active → completed | expired.

Shareable links start as **pending**, activate on conversation create, and terminate as **completed** or **expired**. Completion unlocks transcript persistence and optional intake reporting with `report_sent_at` deduplication.

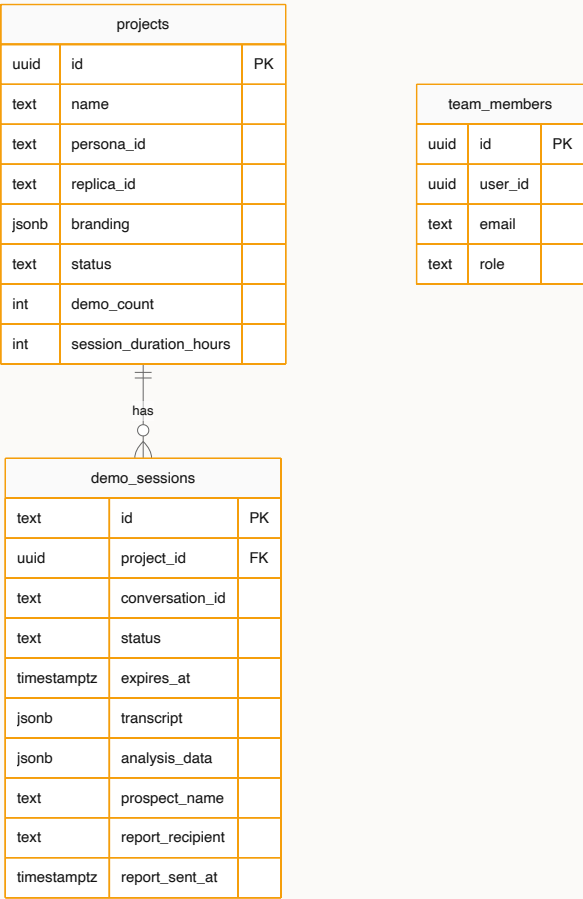
8. Critical end-to-end workflows (continued)



Flo intake sequence: form → Tavus call → webhooks → Claude → PDF → Resend.

Client `POST /api/demo/complete` and `sendBeacon` provide redundancy when webhook timing or transcript readiness is delayed.

9. Data model and state



Core entities: projects, demo_sessions (transcript/analysis/report fields), team_members.

Integrity notes

- Session IDs are custom tokens (demo_*), not UUIDs - shareable URL keys.
- report_sent_at prevents duplicate clinical emails under dual triggers.
- RLS owner policies on projects; public SELECT/UPDATE for pending/active demo sessions.
- Service-role client used for webhooks - elevated privilege path that must stay locked down.

10. Most advanced implementations

1. Clinical intake pipeline (rank 1)

IMPLEMENTED Video interview → transcript acquisition → Claude structured chest-pain schema (CAD/AOD/PE + ROS) → clinical PDF → Resend. Files: `transcript-analyzer.ts`, `pdf-generator.tsx`, `email-service.ts`, `intake-report.ts`.

Difficulty: domain fidelity without hallucination; prompt rules force positive/negative/unknown; clinical shorthand; fail-closed Unknown analysis.

2. Dual-path completion redundancy (rank 2)

IMPLEMENTED Client `/api/demo/complete` after 15s delay, sendBeacon on pagehide, and Tavus webhook `transcription_ready`. Commits `3209aac`, `ac3ace0` show production debugging of event-array transcript extraction.

3. White-label demo control plane (rank 3)

IMPLEMENTED Projects carry branding columns + JSON; 24h sessions; dashboard CRUD (~913 LOC client). Turns Tavus into a deliverable agency product surface.

4. Typed Tavus client (rank 4)

IMPLEMENTED `lib/tavus-client.ts` + `types/tavus.ts` (~461 LOC types). Broader than UI uses (objectives/guardrails/docs/videos) - client breadth is PARTIAL product surface.

10. Most advanced implementations (continued)

5. Supabase schema + RLS + service role (rank 5)

IMPLEMENTED Three tables, enums, indexes, increment_demo_count, transcript migration. Public demo policies carefully allow activation without full admin credentials.

6. Branded demo UX with intake gate (rank 6)

IMPLEMENTED branded-demo-viewer.tsx (~622 LOC): Flo detection by name/flag, form blocking start, report status overlays.

7. Admin project orchestration UI (rank 7)

IMPLEMENTED Persona/replica fetch, branding fields, demo link generation with clipboard fallbacks (multiple hardening commits Jan 2026).

8. Clinical PDF iteration (rank 8)

IMPLEMENTED Commits 1e630bf, 57f96e1 move PDF from generic report to clinical-grade layout.

9. Perception analysis (rank 9)

PARTIAL Captured into analysis_data; rendered in admin send-report PDF path; omitted from automated generateIntakeReport.

10. Multilingual conversation language (rank 10)

IMPLEMENTED Project form + API fields (commits e62485e, 404b6f6).

11. AI system design

This is **application orchestration** over third-party multimodal video (Tavus) and LLM extraction (Claude) - not model training.

LAYER	IMPLEMENTATION	STATUS
Inputs	Live A/V via Tavus replica; pre-call name/email; persona context/greeting	IMPLEMENTED
Orchestration	Session APIs + webhook event routing	IMPLEMENTED
Memory / RAG	No persistent patient memory store; Tavus KB client methods exist unused by Flo path	PARTIAL
Tools	No agent tool-calling loop in-app; structured JSON extraction only	IMPLEMENTED extraction
Outputs	Clinical PDF + email; DB analysis summary	IMPLEMENTED
Evaluation	No eval harness / golden transcripts	PROPOSED
Safety	Prompt anti-hallucination rules; physician review framing in footer	IMPLEMENTED prompt-level
Human control	Doctor is recipient; admin can manually send-report	PARTIAL auth gap on admin route
Cost	Per-call Tavus + Claude + email; no budget instrumentation	PROPOSED

12. Security, privacy, and authorization

- **IMPLEMENTED** Middleware protects `/team/*` except login; dashboard also calls `isAuthenticatedAdmin()`.
- **IMPLEMENTED** Admin email allowlist + shared `ADMIN_PASSWORD` cookie session (7 days, `httpOnly`).
- **IMPLEMENTED** Optional Supabase user acceptance in API helpers (`lib/auth-helpers.ts`).
- **IMPLEMENTED** RLS on tables; service role isolated to webhook/admin ops.
- **GAP** `/api/admin/send-report` has no `requireAuth` - if deployed publicly, anyone who can guess conversation IDs could trigger report generation.
- **PARTIAL** Webhook authenticity verification (signature) not evidenced in handler.
- **INFERRED** Clinical data sensitivity is high; PHI handling depends on deployment practices not fully encoded in repo.

Honesty rule: A login UI is not proof of server enforcement. Enforcement was verified in `middleware.ts`, `lib/auth.ts`, and dashboard page guards - with the admin send-report exception called out.

13. Production readiness and operations

AREA	EVIDENCE	STATUS
Deployment	Vercel-oriented docs; VERCEL_URL webhook fallback	INFERRED
CI	.github/workflows/ci.yml lint/typecheck/build	IMPLEMENTED
Background jobs	No queue workers; webhook + request-time orchestration	IMPLEMENTED sync style
Monitoring	Console logging; no Sentry/Datadog wiring found	PROPOSED
Recovery	Report dedup; dual complete paths; webhook returns 200 on error	PARTIAL
Admin control	Dashboard + manual send-report	PARTIAL

14. Testing and QA evidence

- npm run lint - FAIL - 35 problems (30 errors, 5 warnings)
- npx tsc --noEmit - PASS
- npm run build - PASS (19 routes generated)
- Automated unit/e2e - none; TESTING_CHECKLIST.md is manual
- Portfolio Playwright captures - 14 screenshots under portfolio-showcase/screenshots/

15. Engineering Judgment and Intellectual Work Demonstrated

Evidence-based only. No psychological claims. Confidence: High / Moderate / Limited.

CAPABILITY	EVIDENCE	WHY DEMANDING	VALUE	CONF.
Problem decomposition	Project → session → conversation → report stages	Async multi-system boundary design	Shipable demos	High
Domain modeling	ChestPainIntakeAnalysis schema	Clinical risk taxonomies + anti-hallucination	Physician-usable output	High
Reliability thinking	Webhook + complete + beacon + dedup	Partial failure of transcript timing	Fewer lost reports	High
Security reasoning	RLS + allowlist + middleware	Public demo vs privileged admin	Safe sharing	Moderate
Product reasoning	Branding + 24h links + Flo gate	Agency delivery constraints	Client-ready packaging	High
Debugging under ambiguity	Commits fixing events[] transcript path	Vendor event shape drift	Production recovery	High
Interface/workflow design	Branded viewer + dashboard forms	Non-technical operator UX	Adoption	Moderate
Technical communication	CLAUDE.md, DEMO_WORKFLOW, schema SQL	Multi-doc operational clarity	Handoff	Moderate
Test discipline	0 automated tests	-	Risk	Limited

16. Professional skill matrix

ROLE	WHAT THIS REPO PROVES	GAPS
Senior full-stack	Next.js + Postgres + auth + external APIs end-to-end	Test automation, observability
AI application engineer	LLM structured extraction + multimodal vendor orchestration	Evals, cost controls
Product engineer	Demo ops workflows, branding, clinical packaging	Growth/billing surfaces
Founding engineer	0→1 platform across UI, API, data, integrations	Team process evidence limited to solo git
Solutions architect	Trust boundaries, webhook design, vendor composition	Formal ADRs sparse
Technical lead	Architecture coherence + production fix history	CI green lint; auth gap on admin route

17. Ownership and leadership evidence

IMPLEMENTED Git: Dev Ganuga 42 / Vercel 1 across scaffold → Tavus client → Supabase → branding → Flo clinical pipeline. Byline uses Dev Ganugapenta per portfolio brief. Do not over-claim exclusive commercial ownership beyond commit evidence.

DEPRECATED Early Studio conversation UI removed/redirected (/studio).

INFERRED Create Next App + some docs scaffolding are template-origin; substantive product commits are post-scaffold.

18. Commercial value and defensibility

- Value is in *packaging* Tavus for agency demos and encoding a clinical intake workflow - not in owning the video model.
- Switching cost: project configs, branding, prompt/schema, webhook wiring, operational know-how.
- **UNVERIFIED EXTERNAL CLAIM** Revenue, retention, clinical outcomes - not in repo.

19. Honest limitations and technical risks

- No automated tests - regressions rely on manual checklist.
- ESLint currently failing (35) - CI lint job would fail if enforced on this tree.
- Unauthenticated admin send-report endpoint.
- Perception analysis incomplete on automated PDF path.
- Shared admin password model vs per-user identity for small team.
- Webhook signature verification not evidenced.
- Stale README roadmap still lists items that are now shipped (docs drift).

20. Lessons and future architecture

PROPOSED Authenticate send-report; wire perception into automated PDF; add transcript golden-set evals; queue long report jobs; observability; tighten webhook auth; expand RLS tests.

21. Resume-ready achievements and interview narratives

Technical / product-engineering bullets

- Built ConvoAI Studio on Next.js 16/Supabase to manage Tavus persona-replica demos with 24-hour branded share links.
- Implemented Flo chest-pain intake: transcript → Claude structured extraction → clinical PDF → Resend, with report deduplication.
- Designed dual completion paths (client, sendBeacon, Tavus webhooks) after transcript event-shape production failures.
- Authored typed Tavus API v2 client and conversation lifecycle APIs used by admin and public demo flows.
- Modeled projects/demo_sessions with RLS, branding migrations, and service-role webhook updates.
- Shipped operator dashboard for project CRUD, branding, and demo-link generation with clipboard fallbacks.

Founder / technical-leadership bullets

- Owned 0→1 evolution from Next scaffold to agency demo platform + clinical reporting product surface (42 commits).
- Translated a clinical intake protocol into enforceable LLM schema and physician-facing PDF artifacts.
- Prioritized reliability of async vendor callbacks over single-path happy cases.
- Maintained proprietary NR8R delivery tooling with documented workflows and CI build gates.

Interview narratives (evidence-backed)

1. **Hardest architecture:** Coordinating transcript readiness across iframe end, beacon, and Tavus webhook event types (3209aac).
2. **Reliability:** Deduping report sends when multiple triggers race (report_sent_at).
3. **Product decision:** Encoding Flo intake as first-class project behavior rather than a separate app.
4. **E2E ownership:** Dashboard → link → viewer → Tavus → Claude → PDF → email.
5. **Lesson:** Perception capture without automated PDF wiring shows ship pressure vs completeness - document PARTIAL honestly.

22. One-page summary

ConvoAI Studio is NR8R's internal platform for branded AI video demos and Flo clinical intake reporting - evidence shows real orchestration depth beyond a CRUD wrapper.

Strongest proof
Clinical pipeline + dual-path webhooks + white-label session control plane.

Primary risk
Auth gap on admin send-report; zero automated tests; lint debt.

Stack
Next.js 16, React 19, Supabase, Tavus, Claude, Resend, react-pdf.

Target roles
Founding engineer, AI app engineer, senior full-stack, product engineer.

23. Evidence index (selected)

CLAIM	PATH / SHA
Tavus client	lib/tavus-client.ts
Clinical extraction	lib/transcript-analyzer.ts
PDF	lib/pdf-generator.tsx
Webhook	app/api/webhooks/tavus/route.ts
Dashboard	enhanced-dashboard-client.tsx
Schema	supabase/schema.sql + migrations
Transcript fix	3209aac
Full dossier	portfolio-showcase/TECHNICAL_EVIDENCE.md

End of showcase. Companion one-pager: convoai-studio-executive-one-pager.pdf.